MILITARY SPECIFICATION

LINK, CARTRIDGE, METALLIC BELT, 20MM: M12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of metallic belt link (welded or non-welded) intended to hold 20MM ammunition in the form of flexible belts for use in the M39 type 20MM automatic gun.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-6858	-	Welding, Resistance: Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel/Alloys, Heat-Resisting Alloys, and Titanium Alloys: Spot and Seam
MIL-W-13855	-	Weapons: Small Arms and Aircraft Armament Subsystem General Specification for
DOD-P-16232	-	Phosphate Coatings, Heavy, Manganese or Zinc Base (for Ferrous Metals)
MIL-W-45223	-	Welding, Spot, Hardenable Steels
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168	-	Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.3)

U.S. Army Armament Research & Development Command (ARRADCOM)

Product and Packaging Drawings

7147028	-	Link, Cartridge, Metallic Belt, 20MM, M12
P7147028	-	Packaging Data Sheet for Link, Cartridge, Metallic Belt, 20MM, M12
7147027	-	Body, Inner
7147026	-	Ring, Connector
7145025	-	Bridge, Stress
8415411	-	Bridge, Stress (Alternative)

Inspection Equipment Drawings

7187494	-	Ring Gage, Not Go
7190032	-	Fixture Gage
7190052	-	Plug, Gage, Progressive Flat
7190053	-	Length Gage
7190249	-	Progressive Snap Gage
7190252	-	Plain Snap Gage
7190256	-	Height Gage
7190267	-	Fixture Gage
7190358	-	Pull Test Fixture Gage
7190403	-	Test Fixture Gage
7190404	-	Loading & Unloading Fixture
7190405	-	Steel Plug
7190492	-	Height Gage
7190493	-	Width Gage

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8409284	-	Plate, Special Flat
8415842	-	Ring Gage, Go
8415843	-	Ring Gage, Not Go
8415858	-	Flat Plug Gage
8415859	-	Double End Snap Gage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E18 -	Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B117 -	Method of Test for Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components. The links shall comply with all requirements specified on drawing 7147028 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Pull. Links shall be capable of withstanding 600 + 25 pounds tensile load without fracture. When tensile load is applied, the distance X-X as shown on Drawing 7147028 shall not exceed 1.945 inches at load and the bridge shall be tight on the body after test. Testing shall be as specified in 4.5.1.

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3.4 Twist. Links assembled into belts of 25 links with Government standard 20MM dummy cartridges shall be capable of withstanding a complete spiral twist with no evidence of partial or complete belt separation. Testing shalll be as specified in 4.5.2.

3.5 Cartridge retention. The force required to strip the Government standard 20MM dummy cartridges from links shall be not less than 125 pounds nor more than 300 pounds when applied parallel to the longitudinal centerline of the cartridge. Testing shall be as specified in 4.5.3.

3.6 Free hinging. Links assembled into belts of 25 links each using M51 series 20MM dummy cartridges shall hinge and fold over without binding, kinking, separating or any interference. Testing shall be as spedified in 4.5.4.

3.7 Functioning. The sample of 200 M12 links shall be capable of being assembled into belts with 20MM cartridges and shall operate in the normal firing of an M39 type automatic gun system. The links shall not crack, break, separate or cause a malfunction. Testing shall be as specified in 4.5.5.

3.8 Welding. Welding of the stress bridge shall be in accordance with MIL-W-45223 or Class B of MIL-W-6858. Testing shall be as specified in 4.5.6.

3.9 Marking. The links shall be marked in accordance with Drawing (Dwg) 7147027. In addition, the links shall be identified with the inspection lot number as specified in MIL-STD-1168.

3.10 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship. Workmanship shall be in accordance with MIL-W-13855. Cracks, tears and burrs in non-principal areas and deformations shall not exceed the limits specified in 6.4.1, 6.4.2.2 and 6.4.3.2.

4. QUALITY ASSURANCE PREVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (4.3)
b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Link, Cartridge, Metallic Belt, 20MM: M12	7147028-Sh.l	2113
Link, Cartridge, Metallic Belt, 20MM: M12 (Alternative)	7128028-Sh.2	2000
Body, Inner	7147027	263
Ring, Connector	7147026	263
Bridge, Stress	7147025	200
Bridge, Stress (Alternative)	8415411	263

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. Rejection of the first article sample shall be in accordance with Table I and MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20MM: M12				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ring, Connector - Dwg. 7147026				
	Examination of defects	200	ACC-0 REJ-1	3.2	4.4.2.1
	Decarburization	5	ACC-0 REJ-1	3.2	4.5.7
	Hardness	50	ACC-0 REJ-1	3.2	4.5.8
	Salt spray	8	ACC-0 REJ-1	3.2	4.5.9
	Body, Inner - Dwg. 7147027				
	Examination for defects	200	ACC-0 REJ-1	3.2	4.4.2.2
	Decarburizaton	5	ACC-0 REJ-1	3.2	4.5.7
	Hardness	50	ACC-0 REJ-1	3.2	4.5.8
	Salt spray	8	ACC-0 REJ-1	3.2	4.5.9

NOTES:

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20MM: M12				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Bridge, Stress - Dwg. 7147025				
	Examination for defects	200	ACC-0 REJ-1	3.2	4.4.2.3
	Bridge, Stress (Alternative) - Dwg. 8415411				
	Examination for defects	200	ACC-0 REJ-1	3.2	4.4.2.4
	Decarburization	5	ACC-0 REJ-1	3.2	4.5.7
	Hardness	50	ACC-0 REJ-1	3.2	4.5.8
	Salt spray	8	ACC-0 REJ-1	3.2	4.5.9

NOTES:

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20MM: M12				7147028- Sh 1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Link, Cartridge, Metallic Belt, 20MM: M12				
	Examination for defects	800	ACC-0 REJ-1	3.2	4.4.2.5
	Tests				
	Pull test	200	ACC-1 REJ-2	3.3	4.5.1/7190358 1/
	Twist test	800	ACC-0 REJ-1	3.4	4.5.2 2/
	Cartridge retention	200	ACC-2 REJ-3	3.5	4.5.3/7190403/ 7190404 1/
	Free hinging	800	ACC-0 REJ-1	3.6	4.5.4 2/
	Functioning	200	ACC-0 REJ-1	3.7	4.5.5
	Welding	50	ACC-0 REJ-1	3.8	4.5.6 3/
	Decarburization (body and bridge)	5	ACC-0 REJ-1	3.2	4.5.7
	Hardness (body and bridge)	50	ACC-0 REJ-1	3.2	4.5.8 3/
	Salt spray	8	ACC-0 REJ-1	3.2	4.5.9

NOTES:

1/, 2/, 3/ Tests may be conducted using same sample of links.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20MM: M12 (Alternative)				7147028-Sh 2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Link, Cartridge, Metallic Belt, 20MM: M12 (Alternative)				
	Examination for defects	800	ACC-0 REJ-1	3.2	4.4.2.6
	Tests				
	Pull test	200	ACC-0 REJ-1	3.3	4.5.1/7190358 1/
	Twist test	800	ACC-0 REJ-1	3.4	4.5.2 2/
	Cartridge retention	200	ACC-2 REJ-3	3.5	4.5.3/7190403/ 7190404 1/
	Free Hinging	800	ACC-0 REJ-1	3.6	4.5.4 2/
	Functioning	200	ACC-0 REJ-1	3.7	4.5.5

NOTES:

1/, 2/ Tests may be conducted using same sample of links.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and MIL-W-13855. In addition, inspection lots shall not exceed 500,000 links.

4.4.1.1 Heat treat lot. The heat treat lot shall be limited to one heat of steel and shall consist of links heat treated in a batch type furnace at one change or in a continuous type of furnace during an uninterrupted period extending over not more than one ten hour period.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-12997F(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Ring, Connector		7147026
			NEXT HIGHER ASSEMBLY 7147028

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Cracks-tears or burrs in principle areas		.065%	3.2	Visual/6.4.2.1/6.4.3.1
	Tests:				
102.	Decarburization	5 1/	ACC-0 REJ-1	3.2	4.5.7
103.	Hardness	1/ 3/	.40%	3.2	4.5.8
104.	Salt spray	2/	2/	3.2	4.5.9
MINOR					
201.	Inside diameter		2.50%	3.2	7190052
202.	Incorrect or missing protective finish		2.50%	3.2	Visual
203.	Evidence of poor workmanship		4.00%	3.11	Visual/6.4.1/6.4.2.2/ 6.4.3.3

NOTES:
1/ Selected from each heat treat lot (See 4.4.1.1).
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 7147027
4.4.2.2	Body, Inner				NEXT HIGHER ASSEMBLY 7147028
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Cracks-tears or burrs in principle areas		.065%	3.2	Visual/6.4.2.1/6.4.3.1
102.	Width; tip flare, stress arms (without ring)		.065%	3.2	8415842/8415843
103.	Width, min. opening (grip on cartridge)		.65%	3.2	7190032
104.	Guide tabs offset height; inner surface max; outer surface min. (with steel plug - 7190405)		.65%	3.2	7190256
105.	Missing or improper detents		.65%	3.2	Visual
	Tests				
106.	Decarburization 4/	5 1/	ACC-0 REJ-1	3.2	4.5.7
107.	Hardness 4/	1/ 3/	.40%	3.2	4.5.8
108.	Salt spray 4/	2/	2/	3.2	4.5.9

NOTES: 1/ Selected from each heat treat lot (See 4.4.1.1)
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 7147027
4.4.2.2	Body, Inner				NEXT HIGHER ASSEMBLY 7147028
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR:					
201.	Overall length, guide tabs		1.5%	3.2	7190249
202.	Overall length, of link		1.5%	3.2	7190053
203.	Improper clearance; feeder channel guide		1.5%	3.2	7190267
204.	Overall offset length, max, of guide tabs (with steel plug - 7190405)		1.5%	3.2	7190252
205.	Front surface of ramming guide embossment to rear surface of rear guide tab offset; min.		1.5%	3.2	8415858
206	Front surface of ramming guide embossment to front end of front guide tab (with steel plug - 7190405)		1.5%	3.2	8409284
207.	Missing or improper marking		1.5%	3.9	Visual
208.	Incorrect, missing or defective protective finish		2.5%	3.2	Visual
209.	Evidence of poor workmanship		4.0%	3.11	Visual/6.4.1/6.4.2.2/ 6.4.3.3

NOTES:

4/ Test to be performed for link (alternative) only.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7147025
4.4.2.3	Bridge, Stress 1/				NEXT HIGHER ASSEMBLY 7147028
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101.	Cracks-tears or burrs in principle areas		.065%	3.2	Visual/6.4.2.1/6.4.3.1
MINOR					
201.	Evidence of poor workmanship		4.00%	3.11	Visual/6.4.1/6.4.2.2/ 6.4.3.2

NOTES:
1/ Must meet requirements after assembly in accordance with Dwg. 7147028.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8415411
4.4.2.4	Bridge, Stress (Alternative) 4/				NEXT HIGHER ASSEMBLY 7147028
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101.	Cracks-tears or burrs in principle areas		.065%	3.2	Visual/6.4.2.1/6.4.3.1
102.	Width; tip of stress bridge		.65%	3.2	8415859
	Tests				
103.	Decarburization	5 1/	ACC-0 REJ-1	3.2	4.5.7
104.	Hardness	1/ 3/	.40%	3.2	4.5.8
105.	Salt Spray	2/	2/	3.2	4.5.9
MINOR					
201	Incorrect, missing or defective protective finish		2.50%	3.2	Visual
202.	Evidence of poor workmanship		4.00%	3.11	Visual/6.4.1/6.4.2.2/ 6.4.3.2

NOTES:
1/ Selected from each heat treat lot (4.4.1.1).
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3.
4/ Must meet requirements after assembly in accordance with Dwg. 7147028.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 7147028 - Sht. 1
4.4.2.5	Link, Cartridge, Metallic Belt, 20MM; M12				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101.	Cracks-tears or burrs in principle areas		.065%	3.2	Visual/6.4.2.1 '6.4.3.1
102.	Missing or improper weld buttons		.065%	3.8	Visual/4.5.6
103.	Max., distance between bridge arms or body arms with datum "A"		.25%	3.2	7190493
104.	Min., distance between bridge arms or body arms with datum "A"		.25%	3.2	Gage
105.	Height of bridge tip from bridge base		.65%	3.2	7190492
106.	Width, flared stress arm end (with ring)		1.00%	3.2	7187494
	Tests				
107.	Pull test	200	ACC-2 REJ-3	3.3	4.5.1/7190358 4/
108.	Twist test	800	ACC-0 REJ-1	3.4	4.5.2 5/
109.	Cartridge retention	200	ACC-3 REJ-4	3.5	4.5.3/7190403/ 7190404 4/
110.	Free Hinging	800	ACC-0 REJ-1	3.6	4.5.4 5/

NOTES:
4/, 5/ Tests may be conducted using same sample of links.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 7147028-Sht. 1
4.4.2.5	Link, Cartridge, Metallic Belt, 20MM; M12				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Tests				
111.	Decarburization	5 1/	ACC-0 REJ-1	3.2	4.5.7
112.	Hardness	1/ 3/	.40%	3.2	4.5.8
113.	Salt Spray	2/	2/	3.2	4.5.9
MINOR					
201.	Incorrect, missing or defective protective finish		2.5%	3.2	Visual
202.	Evidence of poor workmanship		4.0%	3.11	Visual/6.4.1/6.4.2.2/ 6.4.3.2

NOTES: 1/ Selected from each heat treat lot (4.4.1.1).
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Link, Cartridge, Metallic Belt, 20MM; M12 (Alternative)				7147028-Sht. 2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101.	Max., distance between bridge arms or body arms with datum "A"		.25%	3.2	7190493
102.	Min., distance between bridge arms or body arms with datum "A"		.25%	3.2	Gage
103.	Height of bridge tip from bridge base		.65%	3.2	7190492
104.	Width, flared stress arm end (with ring)		1.00%	3.2	7187494
	Tests				
105.	Pull test	200	ACC-2 REJ-3	3.3	4.5.1/7190358 1/
106.	Twist test	800	ACC-0 REJ-1	3.4	4.5.2 2/
107.	Cartridge retention	200	ACC-3 REJ-4	3.5	4.5.3/7190403 7190404 1/
108.	Free hinging	800	ACC-0 REJ-1	3.6	4.5.4 2/
MINOR					
201.	Evidence of poor workmanship		4.00%	3.11	Visual/6.4.1/6.4.2.2/ 6.4.3.2

NOTES:
1/, 2/ Tests may be conducted using same sample of links.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-12997F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Link, Cartridge, Metallic Belt, 20MM: M12 Packaging Data Sheet				P7147028
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR	None defined.				
MINOR					
201.	Improper level of packaging and packing		1.50%	3.2	Visual
202.	Incorrect, missing or illegible markings on containers		1.50%	3.2	Visual
203.	Inadequate cleaning or drying of links		1.50%	3.2	Visual
204.	Unit container interior contaminated with dirt, lint, or other foreign matter		1.50%	3.2	Visual
205.	Improper orientation of links in unit container		1.50%	3.2	Visual
206.	Layers of links not separated properly		1.50%	3.2	Visual
207.	Improper closure of unit container		1.50%	3.2	Visual
208.	Improper packing of unit containers		1.50%	3.2	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.2 herein.

4.5 Methods of inspection

4.5.1 Pull test. The link shall be assembled to the test fixture specified herein and the applicable load shall be gradually applied at a uniform rate to that specified on Dwg. 7147028. load shall then be removed and the load and distance X-X, shall be recored and measured. Observation shall be made for compliance with the requirement specified on the applicable drawing. In addition, observation shall be made for link fracture. Upon completion of the test, link so tested shall be scraped.

4.5.2 Twist test. The sample links shall be tested by assembling them into belts of 25 links each using M51 series 20MM dummy cartridges. Each belt shall be laid out horizontally to its full length with the open side of links facing down. Each end of the assembled belt shall be rotated in the opposite directions and continued until belt is completely spiraled. The test shall be repeated with the open side of the links facing up. Observation shall be made for compliance with the applicable specification requirement.

4.5.3 Cartridge retention. The links shall be assembled to the test fixture specified herein and the applicable load shall be gradually applied at a uniform rate to that specified on Dwg. 7147028. The retention force shall be recorded. Partial extraction shall be classified as a defect. Observation shall be made for compliance with the applicable specification requirements.

4.5.4 Free hinging test. The sample links shall be tested by assembling the links with standard 20MM dummy cartridges into belts of 25 linked cartridges. Each belt shall be laid horizontally to its full length, with the open side of the links up, and one end shall be drawn back upon the remainder of the belt until the belt is completely reversed (open side of links down). One end of the reversed belt shall then be drawn back upon the remainder of the belt until the belt is returned to its original (open side of links up) position. Observation shall be made for compliance with the applicable specification requirement.

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4.5.5 Functioning. The sample of 200 links shall be assembled into belts with 20MM cartridges and tested by the Government. The belts shall be fired in bursts of 50 rounds each in an M39 type automatic gun.

4.5.6 Welding. The link samples shall be tested and inspected for welding requirements in accordance with MIL-W-45223 or MIL-W-6858.

4.5.7 Decarburization. The cross section of the sample components shall be polished, etched with 3 to 5% Nital solution and examined under a microscope of not less than 500 power. Observation shall be made for compliance with the applicable drawing requirement.

4.5.8 Hardness. Hardness testing shall be conducted in accordance with ASTM Method E18. Observation shall be made for compliance with the applicable drawing requirement.

4.5.9 Salt spray. Salt spray testing shall be conducted in accordance with ASTM Method B117.Observation shall be made for compliance with the applicable requirements in DOD-P-16232.

5. PACKAGING

5.1 Preservation and packaging. The links shall be preserved and packaged for shipment in accordance with Packaging Data Sheet P7147028 with levels as specified in the contract (See 6.1).

5.2 Packing. The links shall be packed in accordance with Packaging Data Sheet P7147028 with levels as specified in the contract (See 6.1).

5.3 Marking. Marking shall be in accordance with Packaging Data Sheet P7147028.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment designs as required to Commander, AMCCOM, Dover, NJ, 07801, ATTN: DRSMC-QAF-I.

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6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings.Technical data originally prepared by these activities are now under the cognizance of AMCCOM.

6.4 Definition of terms

6.4.1 Deformed link. A link whose shape or dimensions or both have been mutilated or altered by processing to the extent that it is unsuitable for its intended use.

6.4.2 Cracks - tears. See Figures 1 and 2.

6.4.2.1 Cracks - tears in principal areas. Any visible fracture of the link extending through the stock thickness in depth, or a distance in length equal to or greater than the stock thickness as depicted in Figure 1.

6.4.2.2 Cracks - tears in nonprincipal areas. Any visible fracture of the link extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length as depicted in Figure 2.

6.4.3 Burrs

6.4.3.1 Burrs in principal areas. Any rough edge on any part of the link which, in the opinion of the Government rpresentative, would produce a definite removal of metal from a cartridge case or which would injure personnel in handling.

6.4.3.2 Burrs in nonprincipal areas. Any rough edge on any part of the link that does not fall within the description of 6.4.3.1.

6.4.4 Principal areas. The body arms (except for the edges facing inward between the flared tips), stress bridge arms, and the connecting ring.

6.4.5 Partial extraction. When only one end of the mandrel or round becomes free.

6.5 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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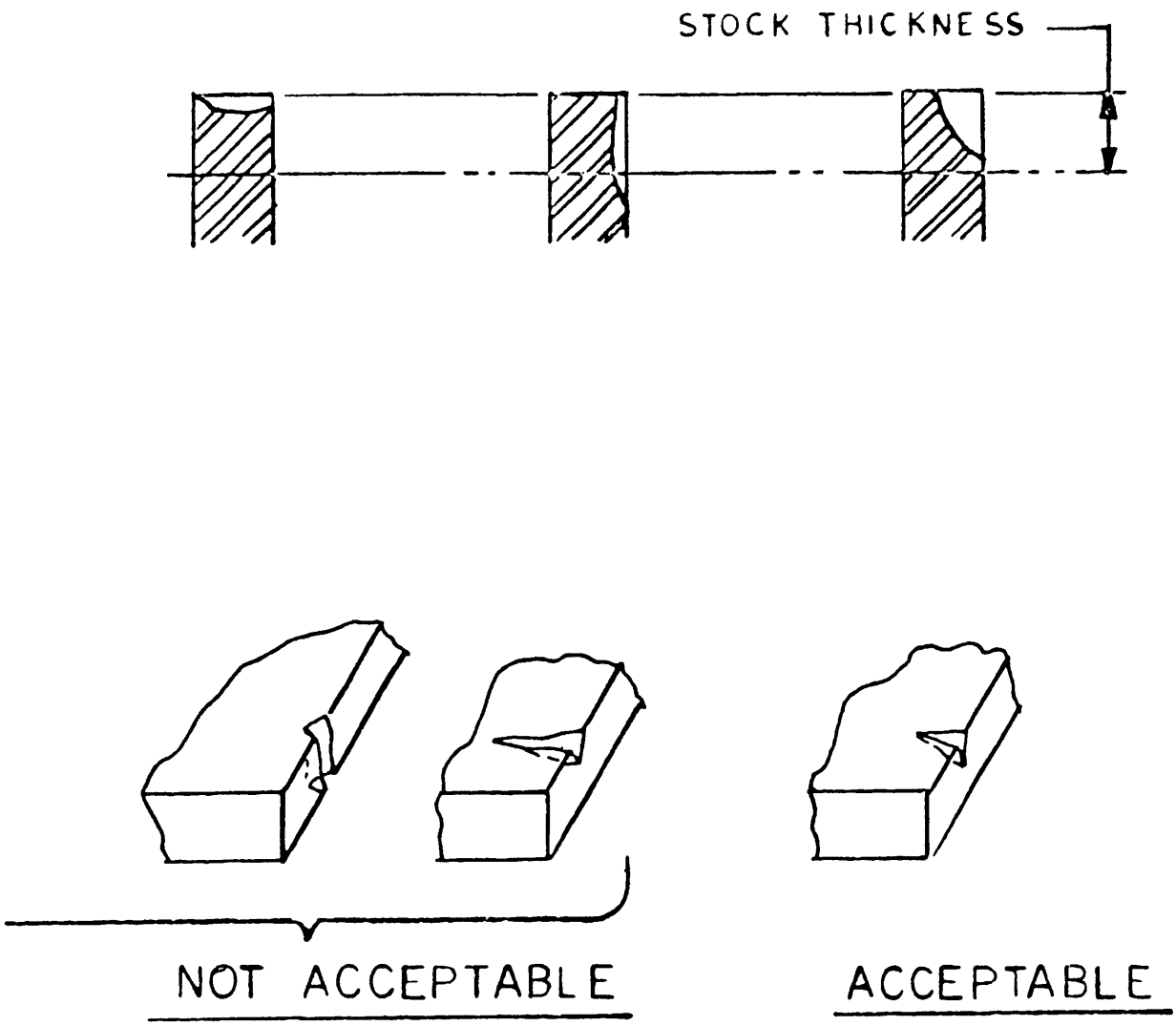
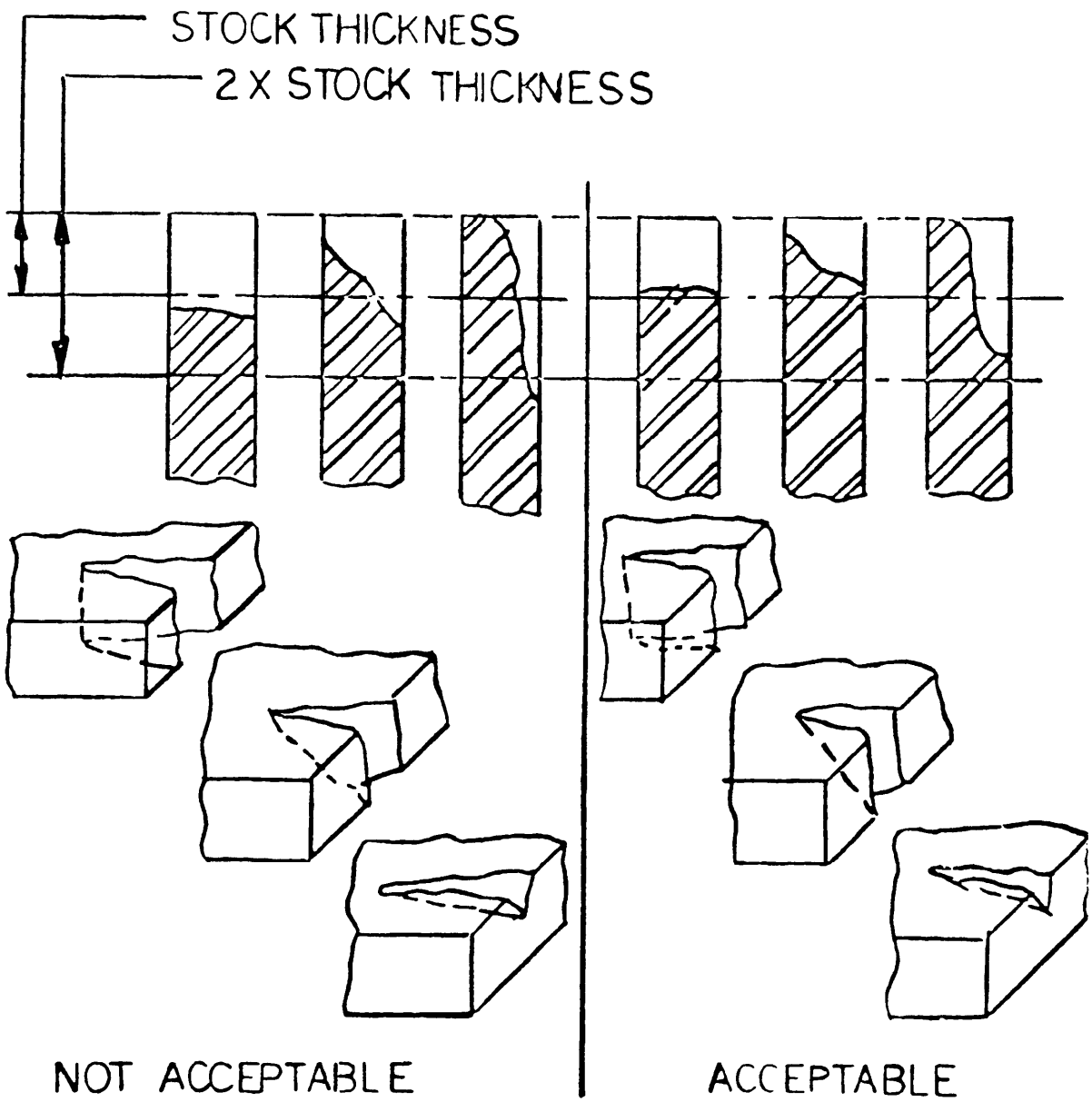


Figure 1

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CRACKS TEARS IN NONPRINCIPAL AREAS

Figure 2

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Review Activities:
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Air Force-99

Preparing Activity:
Army-AR

Project No. 1305-A938

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-12997F	LINK, CARTRIDGE, METALLIC BELT, 20MM: M12

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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